# translationBridgeManagement

## Service Description

Service ID: *"translationBridgeManagement"*

**Abstract**
This document describes the `translationBridgeManagement` service, version 1.0.0, produced by the TranslationManager support system of Arrowhead (documentation edition 5.3.0).

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# 1 Documentation edition and implementation scope

This is the 5.3.0 documentation edition, dated 2026-11-09. It carries forward the system and service contracts from the retained 5.2.2 documentation and identifies the current primary implementation sources. The former `documentation/v5.2.2` folder is retained unchanged.

Document version 5.3.0 is distinct from executable, dependency, registered system/service and wire-contract versions. Values still defined by the source (including 5.2.2 artefact versions, 1.0.0 service versions and `SysMonitor-AMQP-IDD-5.2.2` payload identifiers) are preserved. This documentation update does not change runtime behaviour or establish a new implementation release. Historical verification statements describe the tests recorded in the inherited baseline; this revision validates document compilation and references.

# 2 Overview

This document describes the `translationBridgeManagement` service, which provides management of translation bridges on behalf of consumers: discovery, negotiation, abort and query. It is produced by the TranslationManager system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 3, we describe the abstract message operations provided by the service. In Section 4, we end the document by presenting the data types used by the mentioned operations. The concrete interfaces are specified in the HTTP/TLS/JSON IDD [2] and the MQTT/TLS/JSON IDD [3].

## 2.1 Significant Prior Art

New in Arrowhead 5.

## 2.2 How This Service Is Meant to Be Used

DynamicServiceOrchestration uses `discovery` and `negotiation` on behalf of the consumer during orchestration; operators query and abort bridges.

## 2.3 Important Delimitations and Dependencies

Access follows the management policy of the TranslationManager.

# 3 Service Interface

This section describes the interfaces to the `translationBridgeManagement` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 4.

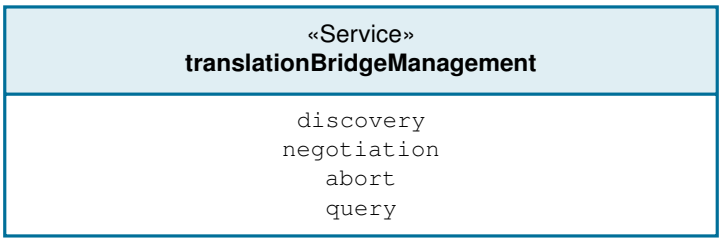



Figure 1: SysML-style block of the `translationBridgeManagement` service and its operations.

The following interface operations are available.

## 3.1 operation discovery (TranslationDiscoveryMgmtRequest) : TranslationDiscoveryResponse

Discovers translation candidates on behalf of the named consumer, with discovery flags.

## 3.2 operation negotiation (TranslationNegotiationMgmtRequest) : TranslationNegotiationResponse

Builds a discovered bridge to the chosen target instance on behalf of its consumer.

## 3.3 operation abort (List<Identifier>) : TranslationAbortMgmtResponse

Aborts the listed bridges; returns the result per bridge.

## 3.4 operation query (TranslationQueryRequest) : TranslationQueryListResponse

Returns a page of bridges matching the filter.

# 4 Information Model

Here, all data objects that can be part of the `translationBridgeManagement` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 4.19, which are used to represent things like hashes and identifiers.

## 4.1 struct Address

A network address with its type.

| Field | Type | Description |
|---|---|---|
| type | String | Address type: `HOSTNAME`, `IPV4`, `IPV6` or `MAC`. |
| address | String | One network address (host name, IPv4, IPv6 or MAC address). |

## 4.2 struct DeviceResponse

A registered device.

| Field | Type | Description |
|---|---|---|
| name | Name | Device name. |
| metadata | Metadata | Free-form metadata (JSON object). |
| addresses | List<Address> | Network addresses with their types. |
| createdAt | DateTime | UTC time of creation. |
| updatedAt | DateTime | UTC time of the last change. |

## 4.3 struct Page

Paging and sorting; all fields optional.

| Field | Type | Description |
|---|---|---|
| page | Number | Zero-based page number. |
| size | Number | Number of records per page (at most `max.page.size`, default 1000). |
| direction | String | Sort direction, `ASC` or `DESC`. |
| sortField | String | Field to sort by. |

## 4.4 struct ServiceDefinitionResponse

A service definition.

| Field | Type | Description |
|---|---|---|
| name | Name | Service definition name. |
| createdAt | DateTime | UTC time of creation. |
| updatedAt | DateTime | UTC time of the last change. |

## 4.5 struct ServiceInstanceInterfaceResponse

One interface of a service instance.

| Field | Type | Description |
|---|---|---|
| templateName | Name | Name of the interface template, snake_case (e.g. `generic_https`). |
| protocol | String | Protocol of the interface, e.g. `http`, `https`, `mqtt`, `mqtts`. |
| policy | String | Security policy of the interface: `NONE`, `CERT_AUTH` or one of the token policies. |
| properties | Map<String, Object> | Interface properties defined by the template, e.g. `accessAddresses`, `accessPort`, `basePath`, `operations`. |

## 4.6 struct ServiceInstanceResponse

A registered service instance.

| Field | Type | Description |
|---|---|---|
| instanceId | Identifier | Service instance identifier `<System>\|<serviceDefinition>\|<version>`, or policy instance identifier. |
| provider | SystemResponse | The provider system. |
| serviceDefinition | ServiceDefinitionResponse | The service definition. |
| version | Version | Version of the service instance. |
| expiresAt | DateTime | UTC expiry time; no value means no expiry. |
| metadata | Metadata | Free-form metadata (JSON object). |
| interfaces | List<ServiceInstanceInterface Response> | Service interfaces offered by the instance. |
| createdAt | DateTime | UTC time of creation. |
| updatedAt | DateTime | UTC time of the last change. |

## 4.7 struct SystemResponse

A registered system.

| Field | Type | Description |
|---|---|---|
| name | Name | System name. |
| metadata | Metadata | Free-form metadata (JSON object). |
| version | Version | Version of the system. |
| addresses | List<Address> | Network addresses with their types. |
| device | DeviceResponse | The device hosting the system, if registered. |
| createdAt | DateTime | UTC time of creation. |
| updatedAt | DateTime | UTC time of the last change. |

## 4.8 struct TranslationAbortMgmtResponse

Results of aborting bridges.

| Field | Type | Description |
|---|---|---|
| results | Map<String, Boolean> | Per bridge identifier, whether it was aborted. |

## 4.9 struct TranslationBridgeCandidate

A candidate reachable through translation.

| Field | Type | Description |
|---|---|---|
| serviceInstanceId | Identifier | Service instance identifier `<System>\|<serviceDefinition>\|<version>`. |
| interfaceTemplateName | String | Interface template of the candidate the bridge would translate to. |

## 4.10 struct TranslationDataModelTranslationDataDescriptor

Settings of a data-model translation.

| Field | Type | Description |
|---|---|---|
| fromModelId | Identifier | Source data model. |
| toModelId | Identifier | Target data model. |
| interfaceProperties | Map<String, Object> | Interface properties of the translator. |
| configurationSettings | Map<String, Object> | Configuration passed to the translator. |

## 4.11 struct TranslationDiscoveryMgmtRequest

Discovery of translation bridges on behalf of a consumer.

| Field | Type | Description |
|---|---|---|
| candidates | List<ServiceInstanceResponse> | Candidate service instances (from service discovery). |
| consumer | Name | Consumer on whose behalf the bridge is discovered. |
| operation | String | QoS operation: `FILTER` or `SORT`. |
| interfaceTemplateNames | List<String> | Accepted interface templates, e.g. `generic_https`. |
| inputDataModelId | Identifier | Data model of the operation input expected by the consumer. |
| outputDataModelId | Identifier | Data model of the operation output expected by the consumer. |
| flags | Map<String, Boolean> | Discovery flags, e.g. `CANDIDATES_AUTH_CHECK`, `CONSUMER_BLACKLIST_CHECK`. |

## 4.12 struct TranslationDiscoveryResponse

Candidates reachable through a translation bridge.

| Field | Type | Description |
|---|---|---|
| bridgeId | Identifier | Identifier (UUID) of the translation bridge. |
| candidates | List<TranslationBridgeCandidate> | Store entries to create. |

## 4.13 struct TranslationInterfaceTranslationDataDescriptor

Settings of an interface translation.

| Field | Type | Description |
|---|---|---|
| fromInterfaceTemplate | String | Interface template of the consumer side. |
| toInterfaceTemplate | String | Interface template of the provider side. |
| token | Token | The token. |
| interfaceProperties | Map<String, Object> | Interface properties of the translator. |
| configurationSettings | Map<String, Object> | Configuration passed to the translator. |

## 4.14 struct TranslationNegotiationMgmtRequest

Builds a discovered bridge on behalf of a consumer.

| Field | Type | Description |
|---|---|---|
| bridgeId | Identifier | Identifier (UUID) of the translation bridge. |
| targetInstanceId | Identifier | Service instance chosen from the discovery candidates. |

## 4.15 struct TranslationNegotiationResponse

The established bridge.

| Field | Type | Description |
|---|---|---|
| bridgeId | Identifier | Identifier (UUID) of the translation bridge. |
| bridgeInterface | ServiceInstanceInterfaceResponse | Interface of the bridge the consumer calls instead of the provider. |
| tokenExpiresAt | DateTime | Expiry of the bridge token. |
| tokenUsageLimit | Number | Number of uses allowed with the bridge token. |

## 4.16 struct TranslationQueryListResponse

List of translation bridges.

| Field | Type | Description |
|---|---|---|
| entries | List<TranslationQueryResponse> | Matching records of the current page. |
| count | Number | Total number of matching records (all pages). |

## 4.17 struct TranslationQueryRequest

Paged filter for translation bridges.

| Field | Type | Description |
|---|---|---|
| pagination | Page | Paging and sorting of the result. |
| bridgeIds | List<Identifier> | Bridge identifiers to match. |
| creators | List<String> | Systems that created the bridges. |
| statuses | List<String> | Bridge statuses to match. |
| consumers | List<Name> | Consumer systems to match. |
| providers | List<Name> | Provider system names to match. |
| serviceDefinitions | List<Name> | Service definition names to match. |
| interfaceTranslators | List<String> | Interface translator systems to match. |
| dataModelTranslators | List<String> | Data model translator systems to match. |
| creationFrom | DateTime | Lower bound of the creation time. |
| creationTo | DateTime | Upper bound of the creation time. |
| alivesFrom | DateTime | Lower bound of the bridge expiry. |
| alivesTo | DateTime | Upper bound of the bridge expiry. |
| minUsage | Number | Minimum number of usage reports. |
| maxUsage | Number | Maximum number of usage reports. |

## 4.18 struct TranslationQueryResponse

A translation bridge.

| Field | Type | Description |
|---|---|---|
| bridgeId | Identifier | Identifier (UUID) of the translation bridge. |
| status | String | Bridge status: `NEW`, `DISCOVERED`, `PENDING`, `INITIALIZED`, `USED`, `ABORTED`, `CLOSED` or `ERROR`. |
| usageReportCount | Number | Number of usage reports received for the bridge. |
| alivesAt | DateTime | Expiry of the bridge. |
| message | String | Text of the log entry or status message of the job. |
| consumer | Name | Name of the consumer system. |
| provider | Name | Name of the provider system. |
| serviceDefinition | Name | Service definition name. |
| operation | String | QoS operation: `FILTER` or `SORT`. |
| interfaceTranslator | String | System translating between the interfaces. |
| interfaceTranslatorData | TranslationInterfaceTranslationDataDescriptor | Settings of the interface translation. |
| inputDataModelTranslator | String | System translating the input data model. |
| inputDataModelTranslatorData | TranslationDataModelTranslationDataDescriptor | Settings of the input data-model translation. |
| outputDataModelTranslator | String | System translating the output data model. |
| outputDataModelTranslatorData | TranslationDataModelTranslationDataDescriptor | Settings of the output data-model translation. |
| createdBy | Name | Name of the system that created the record. |
| createdAt | DateTime | UTC time of creation. |
| updatedAt | DateTime | UTC time of the last change. |

## 4.19 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

| Type | Description |
|---|---|
| Boolean | One out of `true` or `false`. |
| DateTime | Pinpoints a specific moment in time (UTC). |
| Identifier | A string that identifies a record, e.g. a UUID or a composite identifier. |
| List | An *array* of a known number of items, each having type A. |
| Map | A set of key-value pairs. |
| Metadata | A set of key-value pairs with arbitrary, possibly nested values. |
| Name | A string identifier that is intended to be both human and machine-readable. |
| Number | An integer number. |
| Object | Any structured value. |
| String | A chain of characters. |
| Token | An opaque string used as credential. |
| Version | Specifies a version, semantic versioning `major.minor.patch`. |

# 5 References

[1] J. Delsing, “TranslationManager – System Description,” Eclipse Arrowhead, SysD 5.3.0, 2026, ah5-TranslationManager-java-spring/documentation/v5.3.0/TranslationManager-v5.3.0-SysD.pdf.

[2] ——, “translationBridgeManagement HTTP/TLS/JSON – Interface Design Description (TranslationManager),” Eclipse Arrowhead, IDD 5.3.0, 2026, translationManager-v5.3.0-translationBridgeManagement-HTTP-IDD.pdf.

[3] ——, “translationBridgeManagement MQTT/TLS/JSON – Interface Design Description (TranslationManager),” Eclipse Arrowhead, IDD 5.3.0, 2026, translationManager-v5.3.0-translationBridgeManagement-MQTT-IDD.pdf.

# 6 Revision History

## 6.1 Amendments

| No. | Date | Version | Subject of Amendments | Author |
|---|---|---|---|---|
| 1 | 2026-10-22 | 5.2.2 | Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace. | Jerker Delsing |
| 2 | 2026-10-30 | 5.2.2 | Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged. | Claude Opus 5.5 (draft) |
| 3 | 2026-11-09 | 5.3.0 | New documentation edition; primary-source references and Go/Rust implementation scope; 5.2.2 retained. | Documentation maintenance |

## 6.2 Quality Assurance

| No. | Date | Version | Approved by |
|---|---|---|---|
| 1 | | 5.3.0 | |